S2 Supplementary Data

Occupancy tables, LOR join, match support, and discovery local-chart screens

Article type: Supplement

Short title: S2 Data - Apollo propofol occupancy

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S2.0 Occupancy by participant and phase

Full subject table: `results/v1_occupancy_20260921/occupancy_subject.csv`. Cohort medians: baseline 2.71, first sedation 4.07, later sedation 8.68, recovery 3.95. Contrasts versus baseline are reported as Table 1 in the main manuscript (source: `occupancy_contrasts.csv`).

S2.1 Occupancy versus behavioural hit count

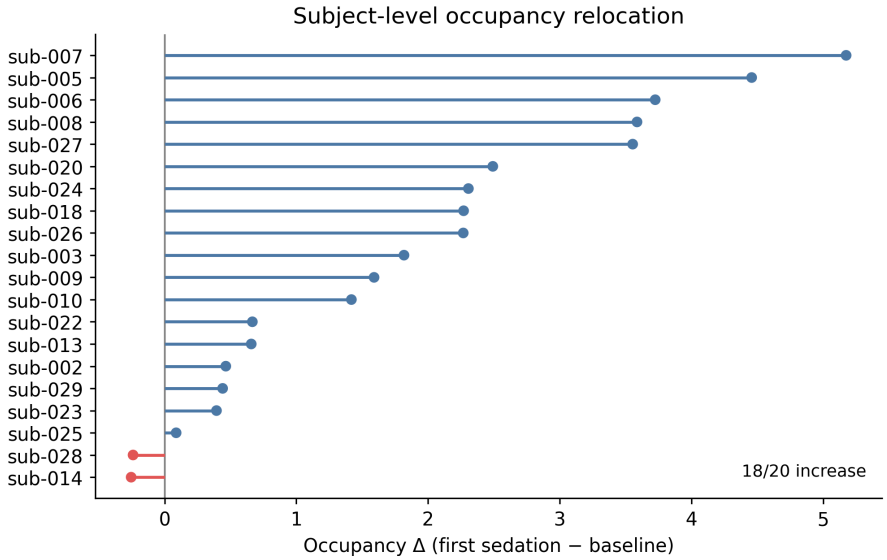


Figure 1: Subject-level baseline-referenced radial-displacement change at first protocol sedation versus baseline. Blue: increase; red: decrease. Source: `results/v1_occupancy_20260921/occupancy_subject.csv`.

The occupancy-versus-hits scatter has moved to the main text because it is the most direct display of the exploratory behavioural question.

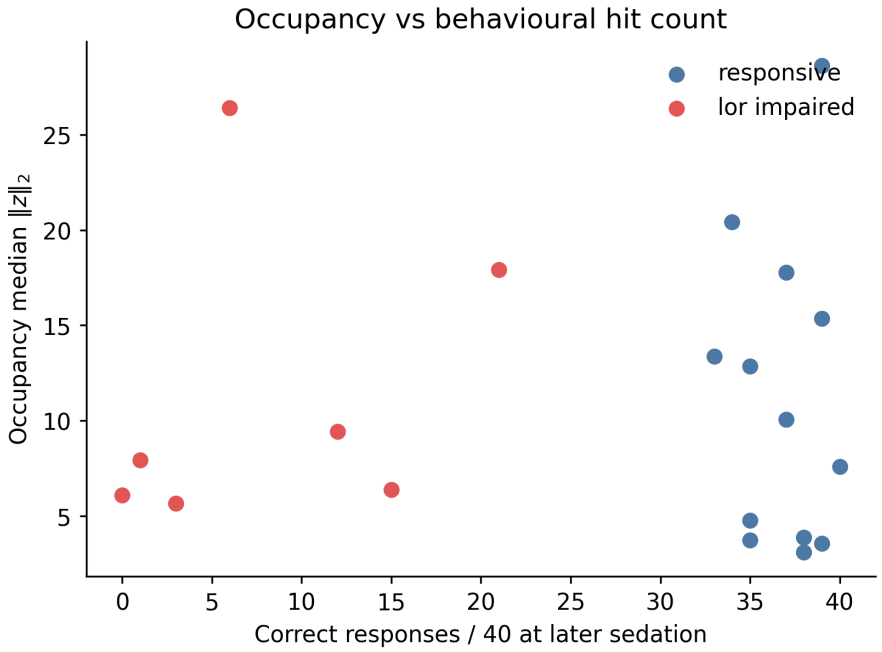


Figure 2: Occupancy versus correct responses / 40 at later protocol sedation. Colour is the occupancy-blind Chennu LOR split. Spearman $\rho = -0.08$, $p = 0.73$, $n = 20$. Source: `results/v1_lor_20260921/lor_occupancy_joined.csv`.

The Spearman estimate is an exploratory descriptive association, not a test of equivalence or independence. The point estimate supplies no evidence of a monotonic occupancy/hit-count association in this $n = 20$ sample, but does not establish that an association is absent; the LOR-group contrast remains NOT_TESTABLE because the impaired arm has $n = 7$.

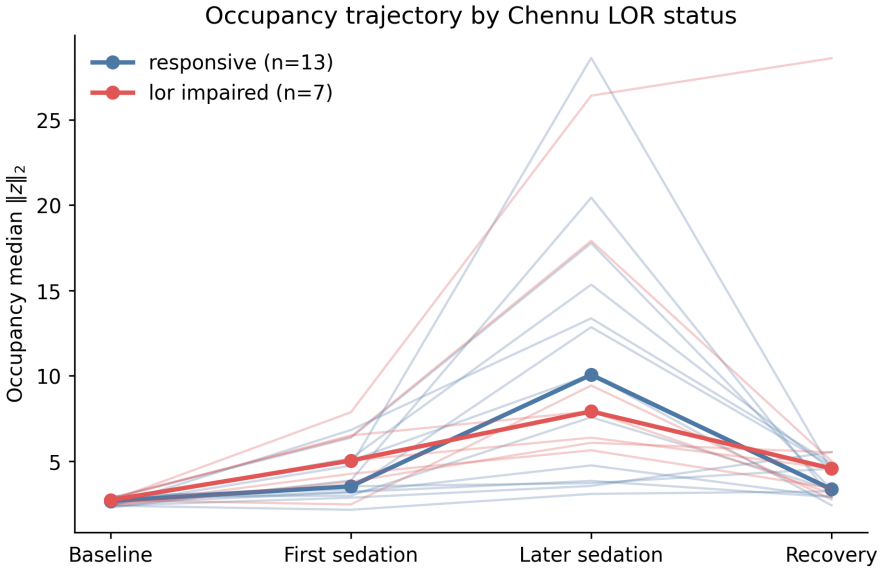


Figure 3: Occupancy trajectory split by occupancy-blind LOR status. Red: LOR-impaired ($n = 7$); blue: responsive ($n = 13$). Medians are within group. This is a descriptive exhibit, not a testable group contrast. Source: `results/v1_lor_20260921/lor_occupancy_joined.csv`.

Hit counts, LOR/ROR status, and concentration: `results/v1_lor_20260921/lor_behaviour_labels.csv`. Group contrasts: `lor_contrasts.csv`. Occupancy-blind split is 7/13, matching Chennu et al. All seven impaired participants returned at recovery.

S2.2 Match support for the baseline-anchor C3 test

Copied from `results/v1_c3_anchor_20260921/` into the occupancy result directory. Included (≥ 20 pairs): sub-002, sub-013, sub-014, sub-022, sub-023, sub-028, sub-029 (7/20). Zero pairs: sub-005, sub-006. Main-text Figure 3 displays this collapse.

On the seven included participants, matched speed $d_z = 0.10$ (Holm $p = 1$) and matched spread $d_z = -0.14$ (Holm $p = 1$). Matched occupancy QC: 2.11 versus 2.16, $d_z = 0.25$. Unmatched increment speed $d_z = +0.31$ and spread $d_z = -0.27$ ($n = 20$) do not reproduce local-chart unmatched spread contraction.

S2.3 Local-chart discovery screen (descriptive)

Exported recording-local dynamical scalars, first protocol sedation minus baseline, $n = 20$ unless noted. Holm applies only to original H1/H2 sign tests. BH-FDR is across the frozen Wilcoxon family. No BH $q < 0.05$. Largest paired $|d_z|$ in this screen: `diffusion_total` $d_z = -0.70$ (sign $p = 0.012$, Holm $p = 0.024$, BH $q = 0.28$). `diffusion_effective_dimension` $d_z = +0.11$. `R_b_over_a` and `A_bD` are empty (0/80). Jacobian spectral abscissa $d_z = -0.31$. Source: `results/v1_stage_20260921/`.

A broader L1/L2/L3 complete-case panel (`results/v1_level_20260921/`) likewise has no BH $q < 0.05$ among tests with $n \geq 10$. Largest L1 $|d_z|$: `chart_drift_b_hat_norm` -0.86 (Wilcoxon $p = 0.0023$, BH $q = 0.24$). L2/L3 complete-case n at first sedation is 4, below the $n = 10$ gate (NOT_TESTABLE). Chart-Jacobian α is Family B, not one-step L3.

S2.4 Occupancy robustness branch

The isolated post-hoc robustness branch is in `results/v1_occupancy_robustness_20260922_reviewfix/`. It keeps the original full-eight-feature robust-z eligibility mask and participant-level pairing. It reports full-MAD occupancy, eight leave-one-feature-out radii, and baseline-only sample-SD and normalised-IQR scale alternatives; the frozen primary outputs are unchanged. All 11 variants retained $n = 20$ paired participants at first sedation. Across variants, the first-sedation median paired difference ranged from 1.295 to 1.757 and d_z from 1.049 to 1.228; the scale alternatives gave $d_z = 1.166$ and 1.171. Feature contributions are descriptive squared-radius shares; beta/alpha ratio and gamma had the largest average shares (18.1% and 16.6%). The lock manifest records exact input, source, plan, and output hashes. This is computational robustness evidence only, not a G3/G4 or biological confirmation.

S2.4a Local-chart matched-state screen (not transported)

S2.5 PR-01 geometric and caliper diagnostics

The separate post-hoc PR-01 branch (`results/v1_pr01_geometric_20260922_final2/`) retained the frozen eight-feature baseline robust-z frame and eligibility mask. At first sedation, phase-cloud centre displacement increased by median 2.60 ($d_z = 1.83$) and within-phase dispersion by 0.477 ($d_z = 0.585$); the 10% covariance-shrinkage-whitened radial displacement gave $d_z = 1.21$. These are descriptive chart-level decompositions, not a latent-state or biological identification.

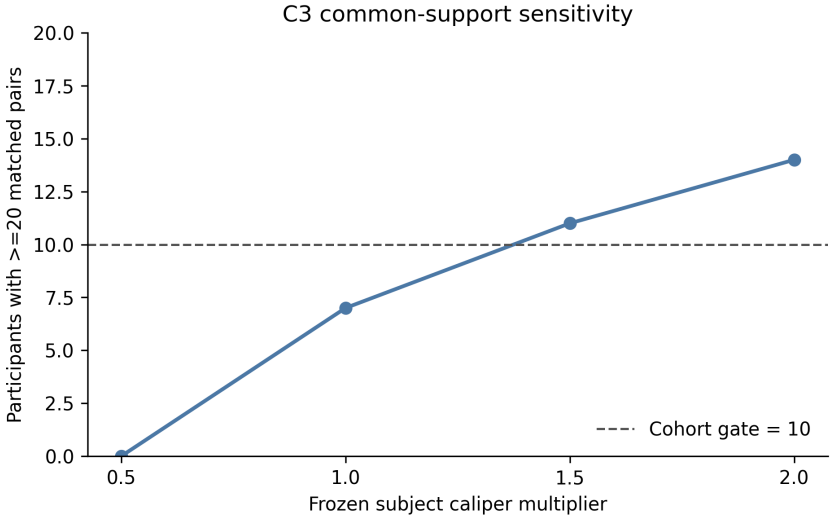


Figure 4: Sensitivity of first-sedation baseline matching support to multiples of the frozen subject-specific caliper. The original 1x gate remains the primary diagnostic; wider calipers are not used to rescue C3. Source: `results/v1_pr01_geometric_20260922_final2/caliper_sensitivity.csv`.

Matching in exported recording-local MNPS coordinates included 19/20 participants. Matched speed $d_z = -0.16$ (Holm $p = 0.36$) and matched spread $d_z = -0.33$ (Holm $p = 0.13$). Unmatched spread $d_z = -0.69$

(sign $p = 0.012$). This chart is re-centred per recording; it is not a baseline-fixed physical frame. Source: `results/v1_c3_20260921/`.

S2.6 Coverage inventory

80/80 files readable. L0/L1 identities finite on 80/80 except `A_bD` and `R_b_over_a` (0/80). Lag-1 one-step L2/L3 finite on 46/80. L4 iterated horizon map 0/80. Destination and resilience `not_testable`. Source: `results/v1_coverage_20260921/`.

Bibliography